Hydrological response of a coastal dune aquifer to plantation clearfell and dune scraping: a BACI analysis of a 21-year manual record

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Abstract

Study region: Newborough Warren, Anglesey, north-west Wales — a single coastal sand body carrying dune-slack habitat of European conservation importance, a mature Corsican pine plantation on the stabilized back dunes, and a retreating Caernarfon Bay shoreline, monitored by a 21-year manual dipwell record (88 wells, 2005–2026).

Study focus: Two restoration interventions are evaluated in a Before–After–Control–Impact design: a 2015 dune scraping at one slack, and an experimental ~4.2 ha clearfell in December 2017. Both are analysed by ANCOVA with a cumulative-water-balance covariate, a distance-weighted scraping term and a coastal-drift control, against five control tiers including in-situ unfelled forest. Effects are reported for the mean water table and summer minimum depth, the metric governing slack viability, then placed on a common site-integrated footing alongside the site's background drivers.

New hydrological insights for the region: After the clearfell the felled well rose +0.128 m over October–March (95% CI [+0.001, +0.254], p = 0.048) and not at all over June–September (−0.001 m); the annual-mean step, +0.108 m, is not significant on its own under autocorrelation-robust (Newey–West) errors (95% CI [−0.021, +0.237], p = 0.10), and summer minima changed at no tier, so canopy removal at this scale does not relieve the summer drought that constrains slack viability. Scraping delivered a larger, directly ecological gain: +0.195 m in summer minimum depth (p = 0.004), but it operates through the same mechanism as coastal erosion, and the drawdown it is predicted to impose on its surroundings, which is modelled and not resolved in the record, carries a chase-inland cascade risk that makes siting decisive. Both interventions are second-order against the site's background drivers: coastal-retreat drawdown, the largest local term, and an unexplained site-wide decline, the largest term integrated over the site.

Highlights

- Felling was followed by a +0.128 m winter rise at the felled well; none in summer.
- Dune scraping lifted the summer minimum +0.195 m — the larger ecological gain.
- Scraping works like coastal erosion; its off-site drawdown is modelled, not seen.
- No felling response was detected outside the forest zone.
- Both interventions are second-order against site-wide and coastal decline.

Keywords: dune slack; groundwater; clearfell; BACI; restoration; coastal dune hydrology

1. Introduction

Coastal dune slacks are among the most species-rich and most threatened groundwater-dependent habitats in north-west Europe. Their characteristic plant communities occupy a narrow band of water-table depth: the seasonal summer minimum must remain shallow enough to wet the rooting zone yet deep enough to avoid permanent inundation, and the community-mean levels of the wet-slack (SD15b) and dry-slack (SD16) communities, means of the four-year average annual minimum, lie roughly 0.4 m apart in Curreli et al. (2013). A complementary indicator based on mean spring rather than summer water level has more recently been proposed (van Willegen et al., 2025); we frame the present analysis on the summer minimum, which is the more direct measure of the drought constraint on slack viability and the metric against which both interventions are evaluated here. Where the summer water table falls progressively, slack vegetation transitions toward dry-grassland and scrub, and the conservation interest is lost. At many British and continental sites this decline has prompted active intervention, most commonly the removal of conifer plantation established on stabilized back dunes during the twentieth century, and the mechanical lowering of the ground surface to bring the water table back into the rooting zone.

The hydrological rationale for these interventions is intuitive but rarely tested against a long record. Plantation removal is expected to recover the rainfall intercepted by the canopy and so to raise recharge; ground-surface lowering is expected to reduce the depth from the surface to the water table directly. Neither expectation, however, distinguishes a change in the mean water level from a change at the summer minimum, and neither accounts for the confounding trends — progressive climate drying and, at coastal sites, shoreline retreat — that operate over the same decades as the management. Without a control-based design and a long baseline, an apparent post-intervention improvement can reflect a wet run of years, and an apparent failure can mask a real benefit eroded by a deteriorating background.

Newborough Warren, on the south-west coast of Anglesey, offers an unusually strong setting for such a test. A continuous manual dipwell record spanning 2005–2026 covers 88 wells across a single sand body, two documented management interventions, and a coastline undergoing measurable retreat. The first intervention is a dune scraping carried out at one slack (CEH36) in April 2015; the second is an experimental clearfell of approximately 4.2 ha of Corsican pine (*Pinus nigra* var. *maritima*) in December 2017, undertaken on the recommendation of earlier researchers and implemented as an operational forestry action. The monitoring network includes wells inside, at the edge of, and well beyond the felled compartment, together with unfelled forest and open-dune controls, allowing the felling signal to be separated from the site-wide climate trajectory and from the coastal-retreat gradient.

This paper analyses both interventions within a Before–After–Control–Impact (BACI) framework and asks three questions. First, did either intervention raise the water table, and if so, in which

season and by how much? Second, what mechanisms explain the response, and what do they imply for the durability of the gain? Third, how far do the effects of forest management reach across the site — in particular, can clearance in the forest zone benefit the eastern slacks where the ecological stakes are highest? The aquifer architecture, cluster structure and state-space model (SSM) on which the analysis rests are developed in the companion paper (Hollingham, 2026a); here we cite that framework rather than re-derive it, and concentrate on the interventions themselves.

2. Study site and interventions

2.1 Site

Newborough Warren is a calcareous coastal dune system of approximately 1,300 ha designated as a Special Area of Conservation for its dune slack habitats. A bedrock ridge forms the northern boundary of the study area; groundwater flows southward from this high ground across a single, highly transmissive body of clean dune sand. The drainage divides around a central dune topographic high: the forested western block discharges south-westward toward Caernarfon Bay, while the open eastern dune discharges south and south-east toward the Menai Strait. This drainage geometry is central to the reach question addressed in Section 5.4: perturbations originating in the western forest propagate seaward, not into the eastern slacks. The geomorphology, the five-cluster partition of the monitoring network and the substrate contrasts are described in full in the companion paper (Hollingham, 2026a).

The monitoring record comprises end-of-month manual dipwell readings. For the analyses reported here the network is partitioned into five hydrogeological clusters — C1 Lake Edge (n = 7), C2 Dune (n = 19), C3 Western Residual (n = 26), C4 Main Forest (n = 9) and C5 Coastal Forest (n = 5), totalling 66 reference wells — with an extended network of additional wells used in robustness checks. The forest occupies clusters C4 and C5; the two carry the same Corsican pine canopy but differ in substrate, C4 sitting on thinner sand over irregular bedrock on the elevated ridge flank and C5 on deeper, more uniform coastal sand.

2.2 The scraping intervention

In April 2015 the ground surface of one dune slack, instrumented by dipwell CEH36, was mechanically lowered to raise the summer water table toward the surface. The excavation depth was not directly surveyed; back-calculation from the observed water-table response and the open-dune specific yield ($d_{ex} = H_0 / Sy$) implies a lowering of approximately 0.43 m. A second scraping episode in October 2023 re-scraped CEH36 and lowered two further slacks (CEH18, CEH21) for the first time. CEH36 lies approximately 340 m seaward and downslope of the later clearfell footprint, a proximity that matters when the interaction between the two interventions is considered. The unscraped slack CEH4, and CEH22 nearer the coast, serve as paired controls.

2.3 The clearfell intervention

In December 2017 approximately 4.2 ha of Corsican pine was felled. The felled compartment straddles the gradational boundary between C4 Main Forest and C5 Coastal Forest, and the wells within and adjacent to it show mixed cluster affinities spanning C3, C4 and C5 (Hollingham, 2026d). The compartment cannot, therefore, be treated as a clean single-cluster experiment; its hydrological effects propagate downslope into the coastal catchment along the continuous drainage pathway. The sole in-situ Impact well is WMC3; four edge wells (CEH31, CEH20, CEH30, CEH16) sit closer to the felling boundary, and unfelled forest, coastal and climate controls complete the five-tier design described below. The dates that define the interventions — the December 2017 clearfell and the dune-scrape years — are the author's recollection, from twenty-one years monitoring the site; the one exception is the April 2015 scrape at CEH36, which is separately documented. No documentary record of the operations has been obtained from NRW or Forestry. These dates set the before/after split of the clearfell BACI.

3. Methods

3.1 BACI design and control tiers

Both interventions are evaluated as monthly BACI gaps — the water level at a target tier minus the level at a control tier — with the intervention "step" estimated as a level shift at the intervention date after adjustment for climate and spatial covariates. The clearfell analysis uses a five-tier classification: Impact (the felled well WMC3), Edge (felled-margin wells), Forest control (unfelled plantation), Coastal control and Climate control. Three counterfactuals are run for the clearfell — Forest, Climate and Combined — but the Forest control is the primary test, because it compares felled wells against unfelled forest sharing the same canopy and substrate and so isolates the canopy-removal response from the coastal-retreat gradient that contaminates comparisons against open-dune controls. The combined control dilutes the felling signal because it incorporates coastal-tier wells that carry the retreat signal, and the climate control spans a wider easting range still; both are reported for completeness but are not interpreted as clearfell effects.

3.2 ANCOVA specification

The clearfell step is estimated by ANCOVA fitted to the monthly BACI gap, with covariates: a centred cumulative water-balance (CWB) term capturing climate forcing; a distance-weighted scraping dummy (exponential decay length L_s = 300 m) that absorbs the 2015 scraping signal at wells near CEH36; a CWB × clearfell interaction testing whether climate sensitivity changed after felling; and a coastal-drift interaction that absorbs progressive divergence between treatment and control wells. The covariate is the fitted coastal-retreat profile averaged over each tier's wells, each at its own distance from the shore, and differenced between target and control, scaled by that differential so its coefficient is dimensionless: a value of one is exactly the amplitude fitted at network scale. It is a drift control, not a measurement of a spatial

gradient. Sensitivity of the clearfell coefficient to the scraping decay length was checked at L_s = 200, 300 and 500 m. Because the monthly residuals are strongly autocorrelated, inference on each clearfell step uses autocorrelation-robust errors (Bartlett kernel, 12 monthly lags; 6 for the October–March fit and 4 for the June–September fit). The scraping intervention is analysed with a two-tier CUSUM-BACI framework: a Tier 1 correction removes background drift estimated at the controls, and a Tier 2 paired comparison estimates the residual scraping signal, cross-checked by synthetic control and by an SSM forward-residual method.

3.3 Coefficient decomposition and the shared SSM

To interpret the BACI steps mechanistically we draw on the state-space model developed in the companion paper, in which the monthly change in water level is partitioned into a rainfall-recharge term (β_1), an atmospheric-draw term (β_2, scaling PET) and a head-dependent drainage term (β_3, scaling displacement above a fixed drainage datum). For the intervention wells the SSM was fitted separately to the pre- and post-felling windows, yielding per-well coefficient shifts ($\Delta\beta_1$, $\Delta\beta_2$, $\Delta\beta_3$) and a predicted clearfell effect computed from those shifts. Canopy interception (24% of incident rainfall; Freeman, 2008) enters the forest clusters implicitly through their lower β_1 and is treated as a partition of the available PET energy budget rather than an additive loss (Hollingham, 2026a).

3.4 Scenario framework

Forward scenarios — full clearfell, 50% thinning and broadleaf conversion — were evaluated by perturbing the per-well β coefficients and computing the sustained change in water level under the no-intercept and free-intercept forms of the SSM (Models A and B; Hollingham, 2026c). Clearfell takes the change in atmospheric draw measured at the felled well, net of the climate controls ($\beta_2 \times 0.952$), with canopy interception unchanged, because the felled well's recharge did not rise after felling; returning the intercepted rain, which a felling wider than the drawdown reach could keep, gives the upper end of the clearfell range, bounded by an open-dune end state; thinning takes half that change, an upper bound, since the remaining trees take up part of the water released (Gebhardt et al., 2014); broadleaf conversion leaves interception unchanged (returning the lower 15% annual-mean deciduous interception of Komatsu et al. (2011) as recharge gives the upper end of its range) and applies a seasonally varying β_2 multiplier that sits below the pine value through the leaf-off winter and above it during the growing season — Nov–Apr and May–Oct means of approximately 0.88 and 1.075 of the pine coefficient, an annual-mean effect close to unity — to reflect deciduous phenology. The framework aggregates per-well responses by inverse-distance weighting; it is a scenario aggregator, not a calibrated continuous-flow model, and applies no explicit propagation between clusters. All numerical results below are read from the live analysis pipeline.

4. Results

4.1 Dune scraping

4.1.1 Background drift

Both scraping controls show accelerating decline independent of any management action. The cumulative CUSUM at CEH4 reached −10.6 m and at CEH22 −19.5 m by early 2026, the convergent deterioration of two spatially separated wells, independent of any management action, consistent with coastal retreat at the western dune edge together with the warren-wide decline, which the record cannot separate, and which any scraping benefit must be assessed against. The summer minimum at the CEH4 control deepened after the pure-scraping era (era means −1.075 m at baseline, −1.053 m during 2015–17, −1.148 m post-felling), indicating the deteriorating background trajectory.

4.1.2 The CEH36 benefit

Three independent methods all return a positive scraping benefit at CEH36 during the pure-scraping era (April 2015 – December 2017). The two control-differenced methods agree closely — a raw BACI step of +0.128 m against CEH4 and a synthetic-control step of +0.141 m from eleven donor wells — while the SSM forward residual, an autonomous-forecast estimator and the most conservative of the three, is lower at +0.100 m (Figure 1). The benefit diminished (to +0.023 m against CEH4) but was not reversed during the post-felling era, as CEH4's own deterioration accounts for much of the apparent reduction in the BACI gap.

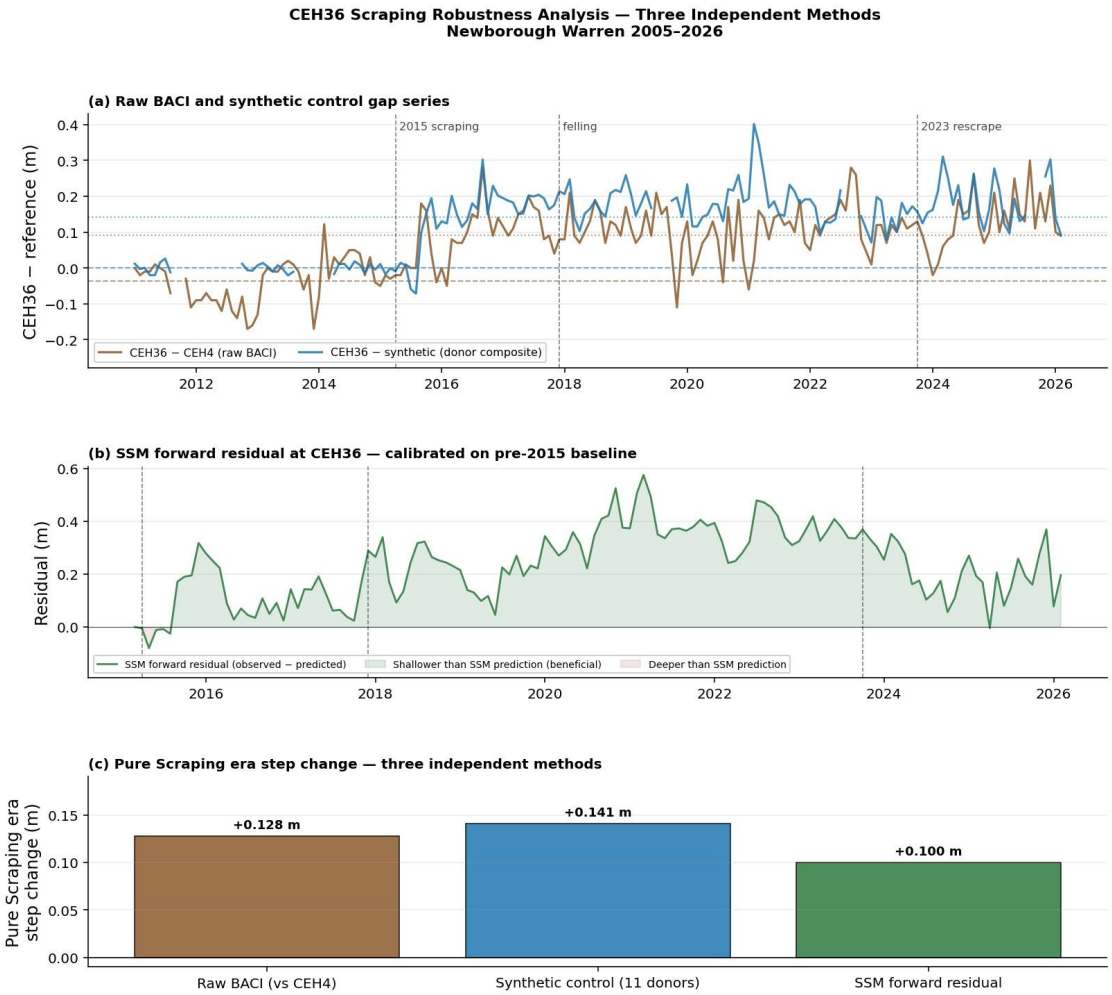


Figure 1. *Three-method robustness assessment of the scraping signal at CEH36 (measured). (a) Raw BACI gap (CEH36 minus CEH4) and synthetic-control gap, with intervention dates; the persistent positive offset after April 2015 marks the step change. (b) SSM forward residual, calibrated on the pre-scraping record; the persistently positive residual indicates a water table sitting above the pre-intervention model prediction. (c) Pure-scraping-era steps from the three methods: raw BACI +0.128 m, synthetic control +0.141 m, SSM residual +0.100 m. Source: 09_scrape_08_ceh36_robustness.*

The ecologically decisive result is at the summer minimum. The paired BACI summer-minimum analysis returns a significant improvement at CEH36 of +0.195 m against CEH4 (p = 0.004) and +0.161 m against the climate-control centroid (p = 0.006); the CEH4 control's own summer minimum deepened after the pure-scraping era (Section 4.1.1), though not significantly against the climate-control centroid (−0.034 m, p = 0.33); the deteriorating background against which the paired gain is measured is evidenced by its Tier 1 CUSUM decline and era means. Neither of the slacks scraped in October 2023 shows a benefit that survives background correction (CEH18 +0.056 m, p = 0.20; CEH21 −0.008 m, p = 0.89), consistent with their short two-summer post-scraping record and, at CEH21, an unreliable control on its own coastal-erosion trajectory.

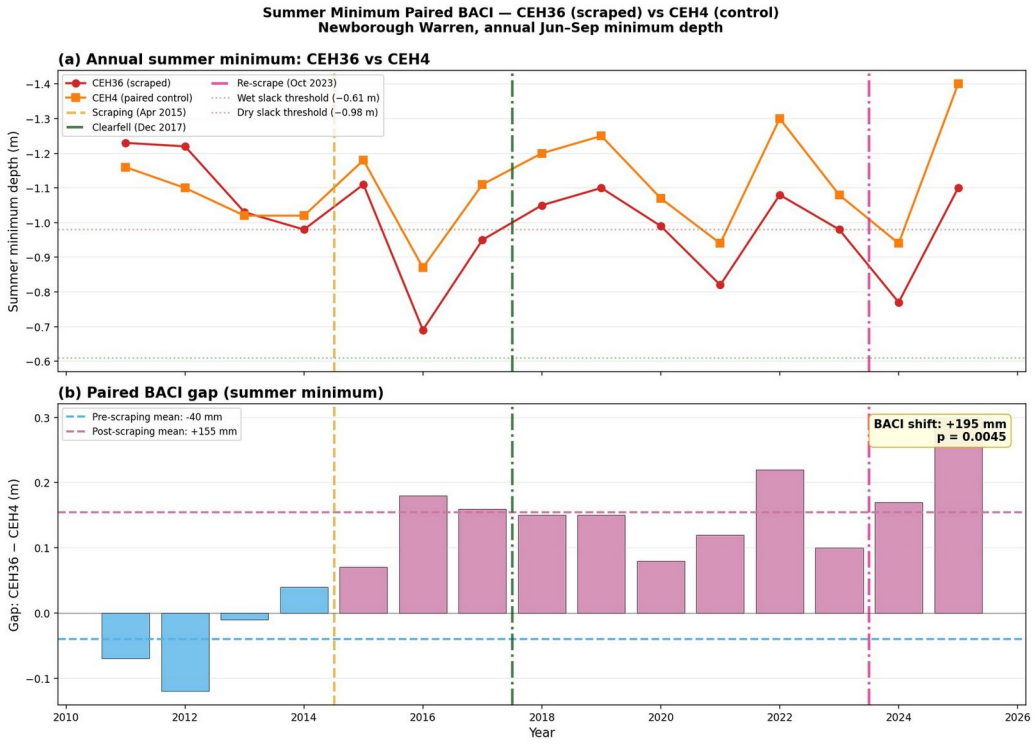


Figure 2. *Paired BACI summer-minimum analysis, CEH36 (scraped) versus CEH4 (control) (measured). (a) Annual summer minimum depth with the Curreli et al. (2013) SD15b and SD16 community-mean levels. (b) BACI gap (CEH36 minus CEH4) by year, pre- and post-scraping means; the paired shift is +0.195 m (p = 0.004). Source: 09c_04_summer_minima_paired.*

4.1.3 Drainage coefficient and propagation

Era-specific drainage coefficients (β_3) increased directionally at both treatment and control wells (CEH36 from 0.084 at baseline to 0.139 during pure scraping; CEH4 from 0.070 to 0.162 over the same period). Because the increase extends to unmanipulated wells, it is not read as a scraping effect: CEH4 rises in the pure-scraping era, before the clearfell, and falls again after it, while the more coastal control CEH22 is flat, and the per-era β_3 values should not be read as evidence of management-driven drainage change. A supplementary test for uphill propagation returned a small positive BACI-corrected $\Delta\beta_3$ at seven of the eight wells within 575 m of CEH36 ($p = 0.043$) but no decay with distance: the uphill wells traverse pre-existing spatial gradients in β_3, which a split-window fit cannot separate from a propagating signal, and the aquifer diffusivity timescale at these distances exceeds the 31-month observation window, so a real drainage front could not have crossed the array within it (Section 5.1.3). No off-site level effect is resolved at the nearest long-record uphill well either. The raw displacement gap at WMC3 — the felled Impact well, some 260 m inland of CEH36 — falls by 46 ± 63 mm at the 2015 scraping and 53 ± 68 mm at the 2023 re-scraping (autocorrelation-robust standard errors); each lies within one standard error of zero, and moving the break date along the record returns a step at least as large at most positions. The drain-cone model predicts about −41 mm at that distance, below what the record can detect: the off-site drawdown is not excluded, but it is modelled,

not measured, and no drawdown cone, near-field or network-wide, should be inferred from the record (Hollingham, 2026d).

4.1.4 Scenario context at the scraped well

To place the measured benefit in context, alternative interventions were computed at CEH36 using its own SSM coefficients. The observed scraping benefit (+46.9 mm water-equivalent per month) substantially exceeds any modelled alternative at the same well: hypothetical clearfell +0.8, 50% thinning +0.4, broadleaf −1.3, and a wet-climate scenario +8.1 mm/month, against −14.3 mm/month under a dry-climate scenario. These modelled alternatives are equilibrium volumetric responses and cannot resolve a summer minimum, so the ecologically decisive comparison rests on the directly observed record: scraping delivered a measured paired-BACI summer-minimum benefit at CEH36 of +0.195 m against CEH4 (Section 4.1.2), a gain none of the modelled forest-management or climate alternatives approaches. The same framework carries an off-site cost that none of the forest-management alternatives carries: a modelled neighbour drawdown of −31.3 mm/month at 100 m, decaying to −16.0 mm/month at 250 m, which the record does not resolve (Section 4.1.3). Whether the on-site benefit works chiefly through the drainage geometry of the cut or through removal of the slack vegetation's growing-season transpiration is not resolved by the record (Hollingham, 2026d).

4.2 The December 2017 clearfell

4.2.1 ANCOVA-BACI results

Against the Forest control — the primary counterfactual — the clearfell step in the mean monthly water table at the Impact well was +0.108 m (95% CI [−0.021, +0.237], p = 0.10, R^2 = 0.23, n = 162), consistent with a recovery but not significant on its own once autocorrelation is allowed for; over October–March alone it was +0.128 m (95% CI [+0.001, +0.254], p = 0.048). The Edge-tier step was small and non-significant (+0.031 m, 95% CI [−0.059, +0.121], p = 0.50). The distance-weighted scraping covariate was significant in both Forest-control models (+0.316 m at Impact, p = 0.019; +0.562 m at Edge, p = 0.002; autocorrelation-robust errors), and clearfell coefficients were identical across the three scraping decay lengths tested, so the scraping and clearfell terms are temporally well separated. The covariate coefficients index that temporal separation in the model; they are not a physical estimate of the scraping effect at the Impact well, and in the climate-corrected fit they carry the opposite sign to the raw scraping-era falls in the WMC3 gap (Section 4.1.3).

The Climate- and Combined-control comparisons are reported in Table 1. The Climate-control comparison returns a null at the Impact well (−0.019 m, p = 0.62) and a fall at the Edge tier (−0.106 m, 95% CI [−0.208, −0.004], p = 0.042). It captures the standing forest's pre-existing hydrological footprint alongside the felling and is not interpreted as a clearfell effect; the Forest control is the appropriate test for canopy-removal attribution (Hollingham, 2026d). The Combined-control estimate (+0.066 m) lies between the two for the same geometric reason. All three comparisons are contaminated to varying degrees by the coastal-retreat gradient, which is why the Forest control is the primary test and Table 1 reports the others for completeness rather than as independent clearfell estimates.

The CWB × clearfell interaction was non-significant in all six models, indicating that the shift that followed canopy removal, significant over October–March but not as an annual mean, left the system's sensitivity to climate forcing unchanged (Figure 3).

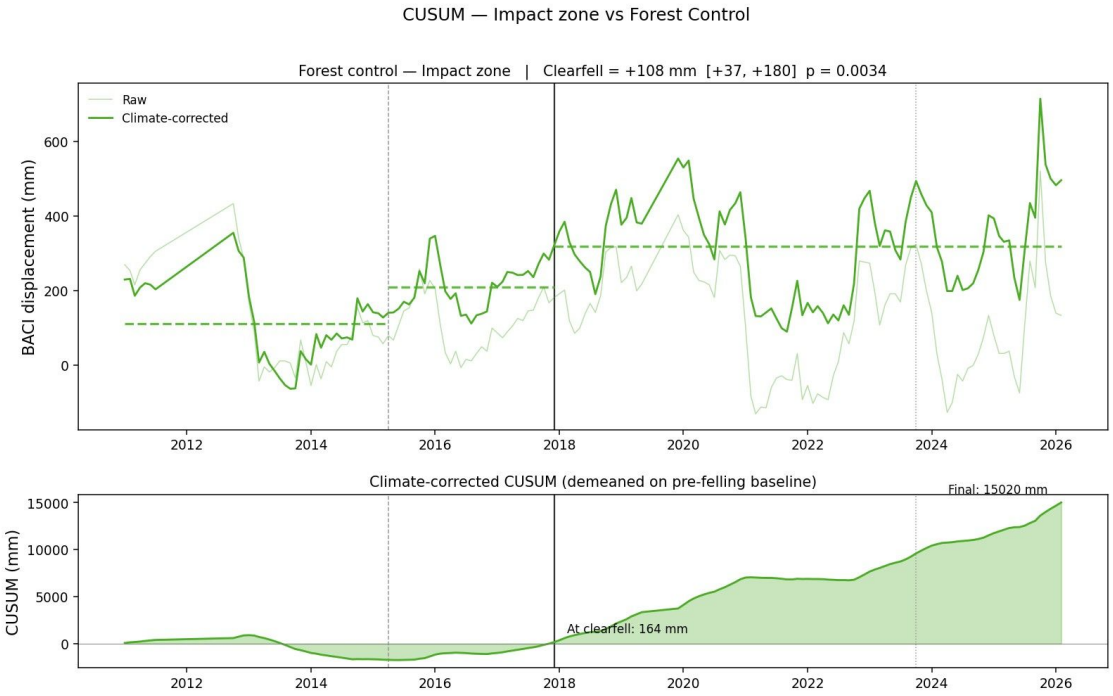


Figure 3. *Forest-control BACI time series and CUSUM for the Impact tier (WMC3) (measured). Upper panel: monthly BACI gap with climate-corrected series and era means. Lower panel: cumulative sum of the climate-corrected series demeaned on the pre-felling baseline; sustained positive accumulation after December 2017 is consistent with the clearfell step. Vertical lines: April 2015 scraping (dashed), December 2017 clearfell (solid), October 2023 re-scraping (dotted). Source: 10a_07_cusum_impact.*

The absorbed differential is not an independent estimate of the coastal gradient: the covariate is that gradient, so setting the two against each other is an internal consistency check rather than corroboration (Hollingham, 2026d). The coastal-retreat premise underlying the correction is independently corroborated by a model-free coast-to-inland transect, in which the coast-minus-inland head difference deepens at −27.8 mm yr^{-1} (95% CI [−36.1, −19.3]), within error of the network-scale gradient anomaly of −31.3 mm yr^{-1} recovered from an entirely independent construction (Hollingham, 2026a). At the Edge tier the drift term absorbs 1.18 ± 0.37 times the network amplitude, likewise not distinguishable from unity; the Edge result is a null with respect to a discrete clearfell step rather than positive evidence of no effect (Hollingham, 2026d).

4.2.2 Summer minima

No significant clearfell effect on summer minimum depth was detected at any tier. At the Impact well the shift was −0.007 m (p = 0.91); the pooled mixed-effects estimate at the Edge tier was −0.064 m and non-significant (p = 0.12). The distributional view by tier (Figure 4) shows

the Impact and Edge tiers tracking the Forest-control trajectory in their summer extremes, with no detectable divergence after felling, while the Coastal control deepens progressively, in a pattern consistent with coastal erosion rather than clearfell timing; its two wells stand 120–180 m from a strip of foreshore pine cleared between October 2014 and March 2015, within the before period, and the comparison carries that clearance as a confound. The clearfell therefore registers in the winter (October–March) level at the felled well, though not significantly as an annual mean, and not in the summer extreme — the metric that determines slack viability — where felled and unfelled wells move together.

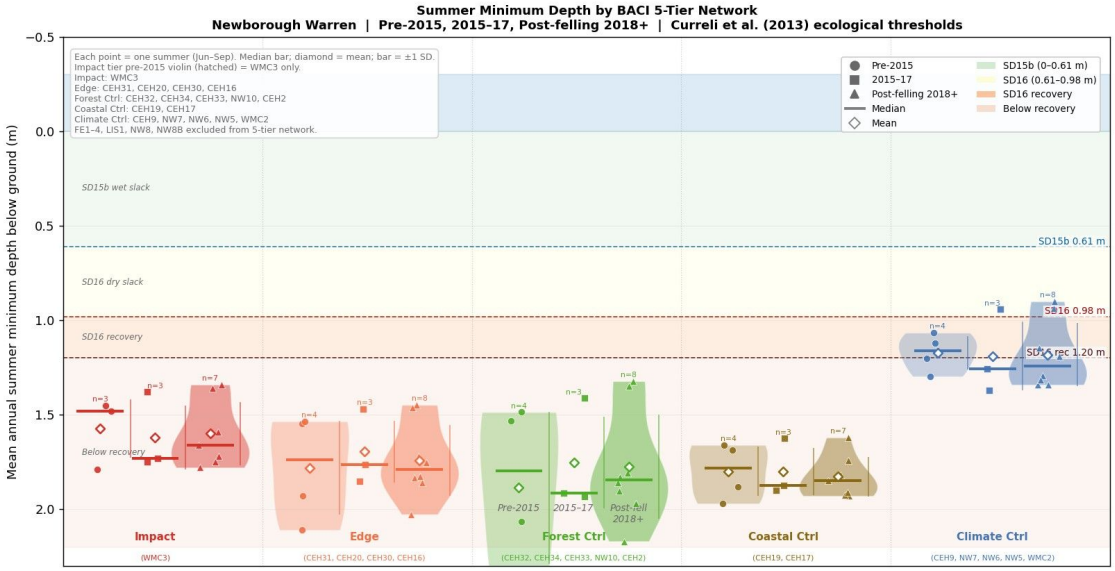


Figure 4. *Summer-minimum depth distributions by BACI tier across three management phases (measured). Violin width indicates frequency; diamond = mean, bar = median. The Impact and Edge tiers track the Forest-control trajectory with no detectable divergence in summer extremes after felling, despite the positive shift (+0.108 m, not significant as an annual mean, p = 0.10) in the monthly ANCOVA; the Coastal control deepens progressively, consistent with coastal erosion, though its wells also carry a 2014–15 foreshore pine clearance as a confound. SD15b and SD16 community-mean levels from Curreli et al. (2013). Source: 21_forestry_04_baci_zone_violin.*

4.2.3 Coefficient decomposition

Fitting the SSM separately to the pre- and post-felling windows shows that recharge sensitivity (β_1) declined at nearly all wells across all tiers — present at the unfelled Forest-control tier as well as the treatment tiers but negligible at the Climate-control tier (−0.3%), so it is unlikely to be clearfell-specific but is not established as independent of management; it is a candidate mechanism for the observed summer-minimum decline (Figure 5). Atmospheric draw (β_2) moved in the opposite direction at the unfelled Corsican-pine Forest-control wells, rising by +0.04 to +0.25 (mean +0.157), whereas it fell at the felled Impact well (−0.27; 4.8% beyond the climate controls), where recharge sensitivity did not rise (0.98 of its pre-felling value net of the climate controls) and at the open-dune Climate and Coastal controls, and the broadleaf well

NW10 fell rather than rose; the β_1 and β_2 shifts are anti-correlated within the Forest-control tier ($r = -0.58$ on absolute shifts, five wells), but that rests on the broadleaf well NW10: the four pine controls alone give $r = +0.17$. Across the 17 BACI-network wells $r = -0.31$, and across all 22 wells, the far-field controls included, -0.74. Drainage (β_3) changes were small and spatially incoherent. Because β_1 falls comparably at controls and felled wells, a per-well reading of the shifts alone would imply no positive clearfell effect, in apparent tension with the positive ANCOVA step.

The two analyses are reconciled rather than contradictory. The ANCOVA measures the gap between felled and control wells, so the site-wide β_1 decline — which affects both — cancels and the canopy-removal effect becomes visible. The per-well decomposition measures coefficient changes at each well individually, where the β_1 decline registers as a negative contribution everywhere and, because controls and felled wells decline comparably, no clearfell-specific signal emerges. The decomposition therefore underestimates the felling benefit; it does not refute it.

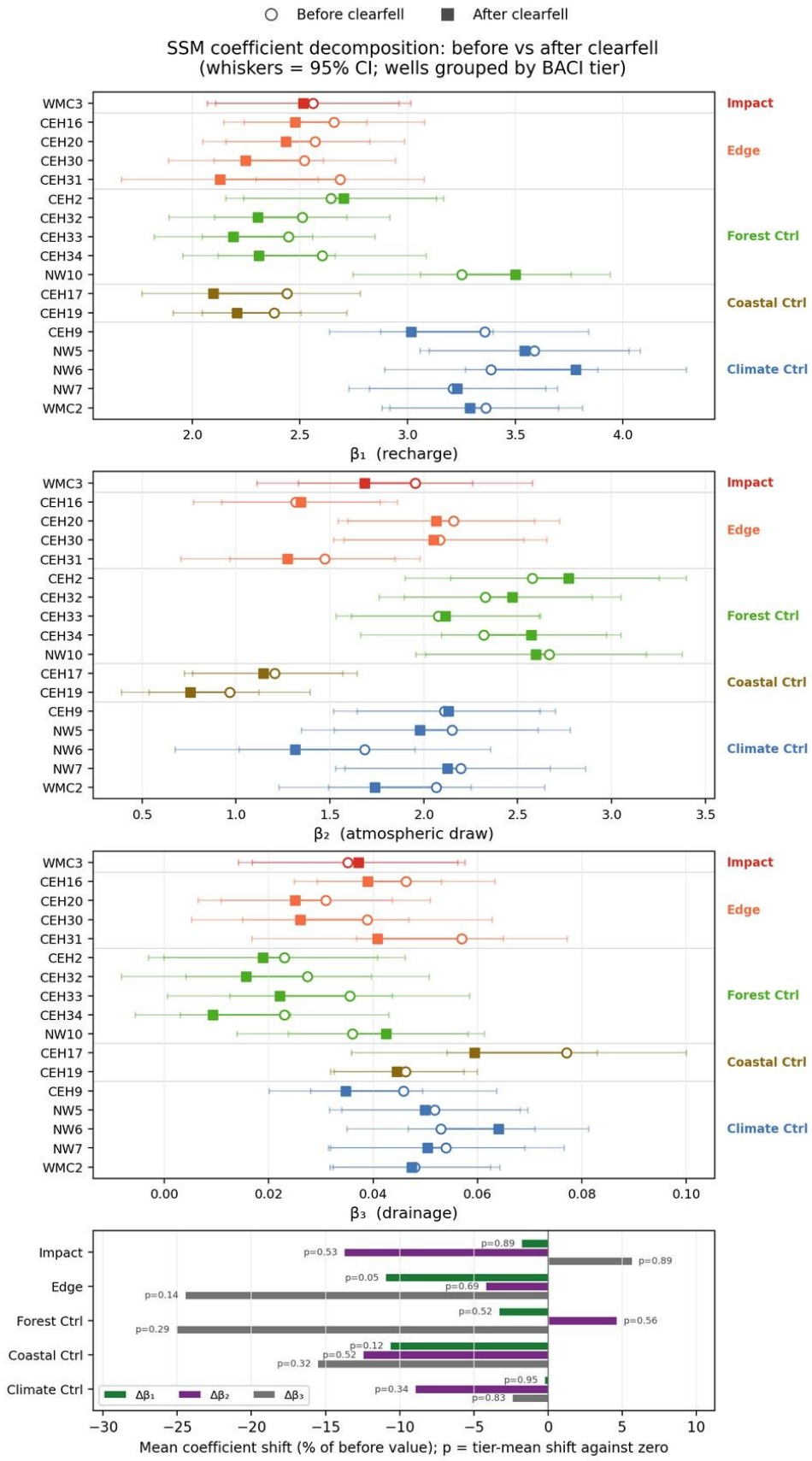

Before clearfell
After clearfell
SSM coefficient decomposition: before vs after clearfell
(whiskers = 95% CI; wells grouped by BACI tier)
WMC3
CEH16
CEH20
CEH30
CEH31
CEH2
CEH32
CEH33
CEH34
NW10
CEH17
CEH19
CEH9
NW5
NW6
NW7
WMC2
Impact
Edge
Forest Ctrl
Coastal Ctrl
Climate Ctrl
β₁ (recharge)
β₂ (atmospheric draw)
β₃ (drainage)
Δβ₁
Δβ₂
Δβ₃
Mean coefficient shift (% of before value); p = tier-mean shift against zero

Figure 5. *Before- and after-clearfell SSM coefficient estimates for the BACI-network wells, grouped and coloured by tier (measured). Open circles = before felling, filled squares = after, joined by a line; whiskers = 95% CI. β_1 recharge declines at most wells, at the unfelled Forest controls as well as the treatment tiers but negligibly at the Climate controls. β_2 atmospheric draw rises only at the unfelled pine Forest controls while falling at the felled Impact well, the open-dune controls and the broadleaf well NW10 — consistent with canopy maturation or with a fitting artefact; the record does not choose between them (§5.3). β_3 drainage changes are small and spatially incoherent. Bottom panel: per-tier mean β_1 and β_2 shift as a percentage of the pre-felling value (β_3 omitted — its proportional swings are noise-dominated and are shown on its own panel above). Source: 10e_03_coefficient_shifts.*

4.2.4 Robustness

Four independent methods relax the ANCOVA assumptions. The six-donor synthetic control returns a clearfell step at the Impact well of +0.101 m (p = 0.002, autocorrelation-robust), estimated against a counterfactual built from donor wells outside the BACI network. The SSM forward-residual estimator, which calibrates each well on its own pre-scraping record and differences the post-felling against the post-scraping era, returns +0.081 m at WMC3 (p < 0.001, autocorrelation-robust). Both are positive and individually significant, but they are not on a common basis and neither is directly comparable to the ANCOVA's control-referenced step. On the SSM-residual estimator the five Forest-control wells drifted upward by a mean of +0.053 m (range +0.016 to +0.124 m), so the Impact-specific component is about +0.027 m: about two-thirds of the felled well's residual is shared with the still-forested controls, and the method attaches no uncertainty to the remainder. The two methods agree that the felled well's gross post-felling step is of order +0.08 to +0.10 m but differ on how much of it survives normalization against the unfelled forest controls, and neither, after that normalization, resolves a clearfell-specific step distinguishable from zero. Synthetic backward extension of the within-compartment wells (WMC3 + FE1 + FE2) returns a clearfell step of +0.078 m (95% CI [−0.016, +0.172], p = 0.10). Decomposed, FE2 — sited inside the felled footprint — diverges +0.029 m post-felling (p < 0.001 with ordinary errors), whereas FE1, some 46 m outside the boundary in standing forest, shows no significant divergence (+0.010 m, p = 0.12): consistent with a within-compartment response, though it rests on two wells and ordinary errors, and no spatial felling footprint is resolved. The WMC3 + FE2 centroid returns +0.094 m (95% CI [−0.024, +0.212], p = 0.12), consistent in size with the WMC3-only step; carrying the same Forest-control structure as the ANCOVA, it is the one direct corroboration of it. A clearfell transect from plantation interior to felling core shows no significant distance gradient, though the post-felling spread in transect hydrographs widens, suggesting a distance-dependent response that seven wells lack the power to resolve. The step is not stationary. Split at January 2021, the Impact-well step against the Forest control falls from +0.152 m (December 2017 to December 2020) to +0.039 m thereafter, no longer distinguishable from zero, and the Edge contrast from +0.066 m to −0.006 m; the decay holds under every coastal-drift treatment tested, and the felled compartment has been kept clear by grazing, so it is not a return of canopy. The +0.108 m is the record mean of a front-loaded recovery (Hollingham, 2026d).

4.3 Forest-management scenarios

The scenario framework evaluates the forest-management scenarios well by well, with no flow between wells, so it cannot show a response at C1, C2 or C3. Full clearfell produces a small positive net annual volumetric response at both forest clusters (C4 +2.8, C5 +1.7 mm/month), the reduction in atmospheric draw measured at the felled well. Fifty per cent thinning gives at most half this response (C4 +1.4, C5 +0.8 mm/month). Broadleaf conversion without an interception gain is small and negative at both clusters (C4 −4.4, C5 −2.6 mm/month), because a canopy in full leaf draws more in summer than a leafless one saves in winter (Figure 6); with a broadleaf canopy's lower interception returned as recharge it would be positive instead, and the record cannot test either end.

C4 captures a larger per-area clearfell response than C5 (+2.8 against +1.7 mm/month), despite carrying the same canopy, because its slower drainage carries the same reduction in draw further. The same canopy treatment therefore delivers different per-area benefit across the forest. These are the lower end of the clearfell range: the 4.2 ha compartment is narrower than the drawdown reach, so a recharge gain there would spread into the surrounding forest, and a wider felling could keep it, bounded by an open-dune end state of 0.5–0.9 m (Hollingham, 2026d). The model still predicts a rise at C5, yet the observed C5 trajectory is the steepest decline in the network; the model-versus-observation discrepancy at C5 is discussed in Section 5.2.3. All three scenario magnitudes are small relative to the climate-driven response over the same horizon, where the dry-climate scenario imposes −12 to −18 mm/month across the clusters.

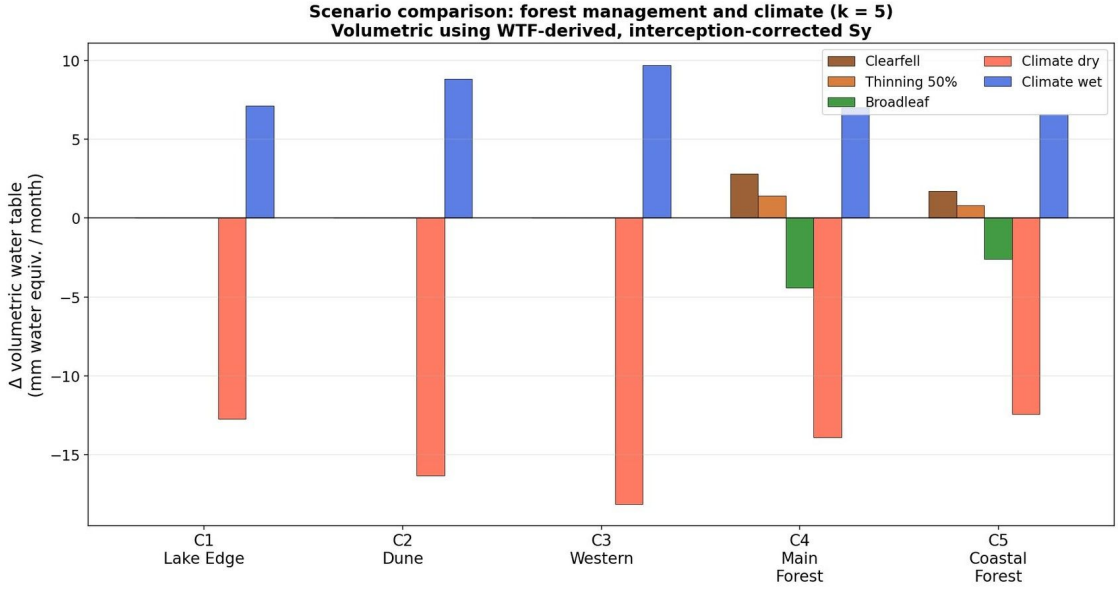


Figure 6. *Forest-management and climate scenario responses by cluster, expressed as volumetric water-equivalent change (mm/month) (modelled). Forest-management scenarios produce a response only at the forested clusters C4 and C5; clearfell and thinning are small and positive, larger at C4; broadleaf conversion without an interception gain is small and negative at both. Climate scenarios are spatially coherent and an order of magnitude larger. Steady-state*

predictions from the per-well SSM coefficient structure. Source: 21_forestry_05_scenario_comparison.

Table 1. Three-counterfactual ANCOVA-BACI clearfell-step estimates (m) against each control tier, with the scraping covariate. The Forest control is the primary test. Climate- and Combined-control rows are contaminated by the coastal-retreat gradient; the Climate-control rows capture the standing forest's pre-existing footprint alongside the felling and are not interpreted as clearfell estimates. The clearfell-step 95% CI and p are autocorrelation-robust (Newey–West); the scraping-covariate p is from ordinary least-squares errors and overstates its significance: on robust errors it is 0.019 and 0.002 in the Forest-control rows (Section 4.2.1) and not significant in the Climate- and Combined-control rows (Combined Edge 0.131, against 0.019 on ordinary errors). Values read from the live pipeline (10a_01_ancova_comparison_table.csv).

Control	Zone	Clearfell step (m)	95% CI, robust	p, robust	Scraping step (m)	Scraping p (OLS)	R^2	n
Forest	Impact	+0.108	[−0.021, +0.237]	0.100	+0.316	0.005	0.232	162
Forest	Edge	+0.031	[−0.059, +0.121]	0.501	+0.562	<0.001	0.455	156
Climate	Impact	−0.019	[−0.097, +0.058]	0.624	+0.014	0.914	0.189	140
Climate	Edge	−0.106	[−0.208, −0.004]	0.042	+0.273	0.513	0.203	136
Combined	Impact	+0.066	[+0.012, +0.119]	0.016	+0.070	0.376	0.259	138
Combined	Edge	−0.021	[−0.068, +0.026]	0.391	+0.300	0.019	0.451	134

5. Discussion

5.1 Dune scraping

5.1.1 Measured benefit and durability

Scraping creates a topographic low that acts as a local drain, raising the water table at the scraped site by collecting lateral inflow. At CEH36 the benefit is clearest at the summer minimum — the +0.195 m paired improvement represents a shift across more than half the gradient separating the SD15b wet-slack and SD16 dry-slack community-mean levels of Curreli et al. (2013) — and it persisted through the post-felling period even as the control deteriorated. Because the benefit is a geometric step rather than a change in dynamic response, the background climate-drying trend continues to operate at the scraped site as elsewhere, and the gain is slowly eroding. The two slacks scraped in 2023 have too short a record, and in one case

too unreliable a control, to support any estimate; a re-scraping interval cannot yet be inferred from the record.

5.1.2 Coastal-erosion analogy and the cascade risk

The mechanism by which scraping delivers its benefit — lowering the ground surface to reduce the depth to water — is physically identical to the mechanism of coastal erosion. Both reduce ground elevation relative to a fixed absolute water-table position, both raise the displacement head at adjacent locations, and both consequently steepen drainage gradients away from the lowered zone. The accelerating CUSUM decline at the two coastal controls is consistent with this process operating naturally at the western shoreline, together with the warren-wide decline, which the record cannot separate. The SSM drainage term operates in both directions, so a scraped zone with elevated displacement head draws water from its surroundings: scraping near the coast steepens the gradient between coast and interior and can deepen the water table at wells uphill.

This carries a management implication the physics make explicit even though the 31-month observation window cannot test it directly. If coastal scraping deepens summer minima at interior wells, those wells may in turn be identified as scraping candidates — lowering their surface, raising their displacement head, and propagating the drainage enhancement further inland. Each successive scrape addresses the local symptom while intensifying the systemic cause. This “chase-inland” dynamic is the topographic analogue of the apparent drainage acceleration that the scraping analysis identifies as a fitting artefact (Section 4.1.3): the aquifer is not gaining storage-decay capacity, but the systematic lowering of ground surfaces produces a head trajectory that the manager may read as progressive deepening requiring further intervention. On the precautionary principle, the spatial planning of any scraping programme should avoid amplifying the existing coastal drainage gradient; the most productive targets are the eastern transitional slacks (Section 5.4), not the coastal margin.

Placing the interventions on the same site-integrated footing as the background drivers characterized in the companion paper gives the cascade a modelled magnitude. A scrape acts as a relief drain: it collects lateral inflow from the surrounding aquifer, and that redistributed inflow is precisely what raises the on-site slack level. The on-site rise is real; in the drain-cone model the redistribution does not net to zero, because the drawdown footprint around each cut depletes more water than the slack recovers. That footprint is modelled throughout: no off-site effect of scraping is resolved in the record, the raw steps at WMC3 lying within its month-to-month variability (Section 4.1.3). On the resulting first-order footing the net effect of scraping, integrated across the site, is a modelled water-table loss of order 25% of the coastal-retreat drawdown in magnitude (Hollingham, 2026d) — a modelled secondary contributor to site-wide decline rather than a local redistribution that balances. This does not diminish the value of a well-sited scrape, whose local summer-minimum gain is measured rather than modelled; it means the benefit is genuinely local and is purchased, on the model, against a modest site-wide cost, which is exactly why siting matters and why the coastal margin — where the off-site drawdown would reinforce the existing erosion gradient — is the wrong target.

5.1.3 Timescale

The propagation timescale is governed by the aquifer's hydraulic diffusivity and, for the relevant distances, is measured in years to decades — longer than the 31-month scraping observation window. This explains why no distance-decay in the drainage signal was detectable uphill of CEH36, and means that whatever coastal-erosion signal the controls now carry is a lagged response to earlier retreat: the present CUSUM trajectories are a leading indicator of inland conditions rather than a real-time measure of them. Monitoring programmes must be designed around this lag.

5.2 The clearfell experiment

5.2.1 A winter-recharge intervention

The clearfell is most directly tested against the in-situ Forest control. After felling, the felled well rose +0.128 m over October–March (95% CI +0.001 to +0.254 m, p = 0.048) and not at all over June–September (−0.001 m); the annual-mean step, +0.108 m, is not significant on its own once autocorrelation is allowed for (95% CI −0.021 to +0.237 m, p = 0.10). Four further estimators relax the ANCOVA's assumptions, but they do not form a converging set: each is built on its own counterfactual and its own reference era, so none is like-for-like with the control-referenced ANCOVA step. Two of them — the six-donor synthetic control (+0.101 m) and the SSM forward residual at the felled well (+0.081 m) — are positive and individually significant; about two-thirds of the SSM residual is shared with the unfelled Forest-control wells (mean +0.053 m), leaving an Impact-specific component of about +0.027 m. The within-compartment backward extension carries the same Forest-control structure as the ANCOVA and is the one direct corroboration of it (WMC3 + FE2, +0.094 m, p = 0.12); the post-felling divergence appears at the well inside the felled footprint (significant with ordinary errors only) and not at the well some 46 m outside it in standing forest, consistent with a within-compartment response, though no spatial felling footprint is resolved across the network. The transect adds no distance gradient that seven wells could resolve.

One sensitivity bears on how the coastal confound enters the step, and its direction is worth stating explicitly. The felled compartment sits closer to the eroding shoreline than the forest controls — about 550 m against 950 m — and the network-scale coastal-retreat gradient predicts a differential of roughly 10 mm yr^{-1} between them. That differential is not left uncorrected: the coastal-drift term in the ANCOVA absorbs it and, fitted freely, does so at 1.51 ± 0.45 times the network amplitude — indistinguishable from feeling it in full (p = 0.25) and firmly distinguishable from feeling none of it (p < 0.001). The +0.108 m step is therefore measured net of the coastal gradient rather than bounded by it; constraining the term to exactly the network amplitude, which the record does not require, lowers the step to +0.077 m (p = 0.002 with ordinary errors), reported as a sensitivity rather than the headline. The step rests on that term: without it the Impact step is +0.016 m (p = 0.51 with ordinary errors), and none of the four robustness estimators carries it, so they and the ANCOVA are not corrected on a common basis. Against the far-field controls the same term returns −1.52 ± 0.61, so the agreement with the network amplitude is not universal across contrasts (Hollingham, 2026d).

The network gradient itself is not applied as a per-well correction: it is fitted with the clearfell wells withheld, accounts for under a fifth of the variance in per-well trend, and individual wells depart from its profile by more than the differential it predicts.

The robustness set does not all point the same way, and the disagreement is reported rather than reconciled: the backward extension corroborates the step directly, while the SSM residual and synthetic control leave no clearfell-specific component distinguishable from zero once the unfelled controls' own drift is removed. The +0.108 m is, moreover, the record mean of an early recovery that is not sustained (Section 4.2.4). The seasonal partition is the firmer result: the felled well rose +0.128 m over October–March ($p = 0.048$), the summer minima show no significant change at any tier, and the step over June–September alone is indistinguishable from zero (−0.001 m, 95% CI −0.118 to +0.116). Canopy removal was followed by a winter rise without relief of the summer drought that governs slack viability. We resist a "success" or "failure" framing: the clearfell was followed by a winter rise and did not do what was hoped, which is lift the summer floor.

The felled well's coefficients do not show the trade between recharge and draw that a sink-and-shield canopy would predict: after felling its recharge sensitivity was 0.98 of its earlier value net of the climate controls, and its atmospheric draw fell, by 4.8% beyond them (Section 5.3). The step is therefore attributed either to that reduced draw or to a recharge gain that the 4.2 ha compartment, narrower than the drawdown reach, does not keep, and the record does not choose between them (Hollingham, 2026d). The non-significant CWB × clearfell interaction is consistent with this: the felling shifted the baseline from which the water table responds to climate, rather than the rate of that response.

An empirical comparison of five-year mean spring water levels (MSL5; van Willegen et al. 2025) between the pre-clearfell baseline (window end 2017, springs 2013–2017) and the current period (window end 2023, springs 2019–2023) taken as a raw difference, shows the network deepening by 0.105 m on average, with the felled well WMC3 (−0.090 m) deepening in line with the network and the Forest-control wells (−0.140 m) somewhat more; against a five-year window standard error of 0.08–0.18 m by cluster, neither departs from the site-wide pattern in a way attributable to the felling. This is consistent with the finding that the clearfell recovery is expressed in the winter monthly level rather than in spring means; the monthly ANCOVA-BACI result remains the estimate of the clearfell step: +0.128 m over October–March (95% CI +0.001 to +0.254 m, $p = 0.048$), the annual-mean step (+0.108 m, 95% CI −0.021 to +0.237 m, $p = 0.10$) not being significant on its own. A window ending 2025 was examined but not adopted: it contains the two wettest springs of the record, and the Forest-control excess it shows is not separable from the spring composition of the window or from the resolution of a five-well group contrast (Hollingham, 2026d).

5.2.2 Reinterpreting the historical baseline

Earlier attributions of water-table depression to the plantation (e.g. Betson et al., 2002) were made during the driest decade in the long RAF Valley rainfall record and without a climate-partitioned, control-based design. The present analysis indicates that the canopy depresses the water table — β_1 is lower under canopy, and the felled well recovered after felling, significantly

over October–March (+0.128 m, p = 0.048) though not as an annual mean (+0.108 m, p = 0.10) — but that the felling left summer extremes unchanged. The historical baseline is best read as a compound of forest influence and exceptional climatic dryness whose seasonal structure was not resolvable at the time.

5.2.3 The C5 model–observation discrepancy

The scenario framework predicts that clearfell raises the C5 water table, through the reduced atmospheric draw measured at the felled well. Yet C5 shows the steepest full-record decline in the network, although the divergence belongs to the early record: refitted from 2011, its summer trend is flat (+0.003 m yr^{-1}, p = 0.79) (Hollingham, 2026d). We flag this discrepancy rather than explain it away. The reconciliation we find most plausible is mechanistic: the clearfell scenario is a *vertical* flux adjustment — a reduction in the canopy's evaporative draw — whereas the observed C5 decline is dominated by a *lateral* geomorphological forcing. C5 sits on deep, freely draining sand directly against the retreating coastal boundary, and its high hydraulic diffusivity allows the lateral drawdown signal from coastal erosion to propagate inland efficiently, overriding the localized vertical gain that the scenario predicts. This does not rescue the prediction so much as locate the missing term: the per-cluster SSM represents vertical exchange but not the lateral coastal boundary condition, and the mixed felling zone straddling the C4/C5 transition compounds the attribution. The coastal-retreat signal is the leading candidate. The network-scale gradient analysis places it on a quantitative footing: progressive coastal-boundary retreat accounts for approximately 53% of C5's observed summer-minimum decline, leaving −17.6 mm yr^{-1} unexplained, comparable to the −13.0 and −17.2 mm yr^{-1} left unexplained at C1 and C2, which carry little or no coastal term (C3, half of whose wells lie within the reach, leaves −3.2 mm yr^{-1}); the thin-saturated-thickness and canopy-maturation mechanisms are not excluded and retain room for a quantitative role, and the attribution is sensitive to which wells the cluster contains; the common-mode background contributes a small term of the opposite sign that the fit does not separately identify (Hollingham, 2026d). The discrepancy is therefore partly reconciled: the scenario framework predicts the change a felling would add, not the cluster's trajectory, and C5 carries a structural coastal-retreat component that clearfell cannot offset, so the predicted gain addresses the climatic component only (Hollingham, 2026d). The precise 53% is fragile to which C5 well anchors the forest-free gradient fit, but the direction is robust — C5 is the most coastal-affected cluster on this fit — and continued C5 monitoring with targeted coastal-retreat measurement remains a stated research priority.

5.3 Hydrological role of the plantation

The forest functions as both a sink and a shield. As a sink it intercepts about a quarter of the rainfall (Freeman, 2008) and draws the water table down through transpiration; as a shield it takes up evaporative energy at the canopy before that energy reaches the water table. The cross-cluster β_2 contrast is dominated by substrate rather than canopy. C4 carries the network's highest β_2 not because its canopy draws hardest but because its thin sand over irregular bedrock has the lowest effective specific yield, so a given evaporative flux produces the largest head change; C5, on deeper sand, has a much lower β_2 despite identical canopy (Hollingham,

2026a). The practical consequence is that canopy-attributable changes in β_2 must be measured against a position-matched control — precisely the role of the Forest-control tier here.

The felling tests both roles, and neither appears at the felled well as the canopy framework predicted. Had the canopy acted mainly as a sink, its removal should have returned the intercepted rain as recharge; yet WMC3's recharge sensitivity after felling was 0.98 of its earlier value, net of the climate controls (standard error 0.13), where a full local return would have raised it to 1.32. Had it acted mainly as a shield, the exposed ground should have dried faster; yet atmospheric draw fell after felling, by 4.8% beyond the controls, and by 1.7% in the canopy-on months. Two readings are consistent with the record, and it does not choose between them. In the first, felling acted chiefly by reducing the draw, and the scenario framework is built on that measured change (Section 4.3). In the second, it did return recharge, but the 4.2 ha compartment, a disc of radius about 115 m, is narrower than the drawdown reach of about 220 m, so the gain spreads into the surrounding forest: a disc of that size keeps about 15% of a full local gain at WMC3 (7–28% across the range of conductivity), a rise in β_1 the record cannot distinguish from none. The Edge wells would receive about half that share, and they rose +0.031 m, not significantly. The record therefore rules out a full local gain but is consistent with one that has dispersed; grazing kept the compartment open, so regrowth does not explain the difference. Field studies of clearcutting report rises of the order of the measured step: 0.13 m on wet mineral soils, where interception fell from 31% to 10% of rainfall (Roy et al., 2000), and 0.29 m in pine flatwood uplands when the whole area was harvested, with no significant upland rise when only the wetlands within it were cut (Sun et al., 2000). A felling wider than the drawdown reach could keep the gain this one did not, so returning the intercepted rain marks the upper end of the clearfell range, bounded by an open-dune end state, rather than its central estimate. Nor has felling undone the plantation's long-run effect: with open-dune coefficients the forest clusters would stand some 0.5–0.9 m higher, whereas the felled well's recovery is about a tenth of a metre (Hollingham, 2026d).

That control tier behaved differently: atmospheric draw rose at the unfelled pine controls even as it fell almost everywhere else, and refitting β_2 on a common PET support across the two eras leaves the rise in place, so a change in climate forcing common to all tiers does not account for it. The shift is specific to the pine controls — the leaf-off broadleaf well moved the other way and the open-dune controls fell. Two readings are possible, and the record does not choose between them. The rise is consistent with canopy maturation (Hollingham, 2026d); two of the five controls sit inside a 1998 replant block whose canopy was closing across the monitoring window. It is also consistent with an artefact of fitting: the rise is modest, not significant well by well, and within the pine controls it is mirrored on average by the β_1 decline, the pattern an ordinary-least-squares fit trading variance between two seasonally anti-correlated regressors — winter rainfall against summer evaporative demand — would leave at wells where the canopy buffers and lags recharge enough to make the two hard to separate. Well by well, however, the two shifts are not anti-correlated across the four pine controls ($r = +0.17$), so this reading rests on the tier means rather than on the per-well signature.

A warmer-winter increase in canopy interception, physically reasonable for an evergreen stand, is consistent with the recharge (β_1) decline but cannot account for the β_2 rise. In the

Thornthwaite scheme used here winter PET is small and essentially unchanged between the two windows — the entire PET increase falls in summer — and the scheme cannot represent the wind- and humidity-driven evaporation of intercepted water at all, so increased winter interception would register as suppressed recharge, not as stronger atmospheric draw. The remaining physical possibility is a summer storage effect — as the table declines into the thinner, lower-specific-yield C4 sand, a given evaporative flux produces a larger head change, inflating apparent β_2 — but this too is a property of substrate and a falling table, not of the canopy. We therefore do not read the Forest-control β_2 rise as a measured change in forest water use; consistent with the companion paper's finding that absolute β_2 values are upper bounds best read as rankings (Hollingham, 2026a), the decomposition is reliable for the direction of the β_1 decline but not for a quantitative reading of β_2.

A direct observational test of broadleaf conversion exists within the network, and it is best read as an indicative phenological comparison rather than a spatial counterfactual. Dipwell NW10 sits in a block of mixed broadleaf established after a 1993 clearance. Its decoupled SSM coefficients capture the deciduous seasonal partition directly: a higher recharge sensitivity (β_1 = 3.48 against a C4 median of 2.33), consistent with a leaf-off canopy transmitting more winter rainfall, set against an atmospheric draw (β_2 = 2.72) comparable to adjacent pine, consistent with that winter advantage being offset by growing-season transpiration. Its climate-normalized summer minimum sits about +0.25 m shallower than the pine interior, and the anomaly declines over the post-felling summers (−26 mm yr^{-1}), in the direction the lower end of the broadleaf scenario range implies, though the trend is not significant ($p = 0.10$) and supports no extrapolation. The spatial reading is weak: NW10 is a single, positionally confounded well, and the spread between two pine wells in the same cluster exceeds the NW10 anomaly, so it supports an indicative reading at most — a maturing broadleaf canopy whose winter leaf-off advantage is partly given back in summer. A robust species attribution would require several instrumented wells across the broadleaf block.

We also note a slower process that the water-table analysis cannot resolve but that bears on restoration: progressive *Dothistroma* needle blight is already thinning parts of the canopy, offering a natural experiment in canopy-density effects on β_2 that requires only sustained monitoring. It is flagged as a monitoring question rather than a finding.

5.4 The reach of forest management

A central management question is whether forest management in the C4/C5 zone could improve conditions in the eastern open-dune clusters (C1, C2, C3), where summer minima sit closest to the SD15b and SD16 community-mean levels and the remaining wet-slack resource is concentrated. The evidence indicates it cannot. The scenario framework evaluates forest management well by well, with no flow between wells, so it cannot itself show a response at C1–C3; the BACI record shows no felling-specific response at wells outside the forest zone once common-mode climate variability is removed; and the post-felling transect is spatially uniform with no inland gradient. This isolation is not a framework artefact. The drainage geometry sends forest-zone perturbations south-westward toward Caernarfon Bay, not eastward into C1 and C2 (Figure 7), and the modelled steady-state drawdown of the existing plantation has an e-

folding length of about 220 m — an upper bound on its reach beyond the canopy, though, carrying interception alone, only a lower bound on its magnitude — reaching the lake-edge wells nearest the forest at 8–72 mm and the eastern dune and lake-edge wells furthest from it at 2 mm or less (Hollingham, 2026d). A felling narrower than that reach does not keep a local recharge gain, which spreads into the surrounding forest; the scenarios cannot show how far a recovery would travel.

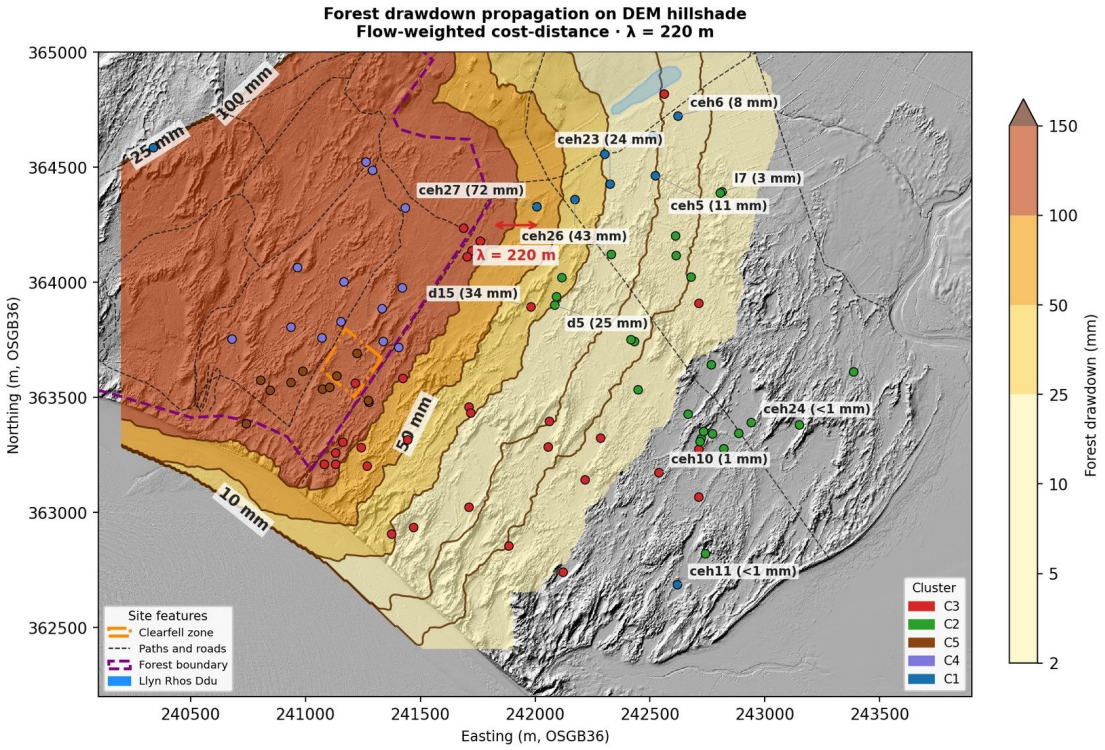


Figure 7. *Steady-state drawdown field of the existing plantation, illustrating the reach of forest-zone perturbations (illustrative — computed from an assumed canopy-edge head deficit, not a measured drawdown). The perturbation is directed south-westward toward Caernarfon Bay along the drainage gradient and falls to 8–72 mm at the lake-edge wells nearest the forest and to 2 mm or less at the eastern wells furthest from it. The per-well inverse-distance framework is a geometric aggregator, not a calibrated flow model. Source: 20_drawdown_propagation.*

The clusters where the stakes are highest — C1 and C2, whose record-mean summer minima sit 0.240 m and 0.325 m below the SD15b community-mean level and above the SD16 community mean — therefore show no detectable response to forest management, and their summer trajectory is set by the site-wide climate signal. Peak winter level is predicted by winter rainfall at C1 and C2, where the antecedent summer minimum is not significant; at C3 the two carry equal weight ($p = 0.047$ and 0.045); and at C4 and C5 antecedent summer depth dominates ($p < 0.001$) (Hollingham, 2026d). Summer minimum depth is therefore the restoration lever in the forest clusters, shared with winter rainfall at C3; over the summer minima at C1 and C2 the clearfell BACI detected no influence of forest management in the C4/C5 zone, though the

model implies a very small one: the existing plantation's drawdown reaches 8–72 mm at the lake-edge wells nearest the forest and 2 mm or less further east (Figure 7).

5.5 Implications for restoration and monitoring

Three conclusions for management follow. First, summer minimum depth is the binding ecological constraint and, in the forest clusters, the restoration lever, shared with winter rainfall at C3 (at C1 and C2 winter flooding follows winter rainfall); there, summer dipwell readings are more diagnostically informative than winter peaks, which reframes monitoring design accordingly. Second, topographic scraping is the most directly effective intervention in the record, but its benefit is local and its planning must respect the cascade risk by targeting the eastern C1/C2 transitional slacks — where a single scrape can bring a marginal well within reach of a mildly wet winter — rather than the coastal margin where it would amplify an existing erosion-driven gradient (Figure 8). Third, plantation clearance at this scale was followed by a winter rise at the felled well but not by a rise over the summer months or in the summer floor, and no felling response was detected outside the forest zone; its hydrological case rests on within-forest objectives, not on benefiting the wider slack resource.

Two points follow for monitoring design. First, the aquifer equilibrates to a step change over years rather than months — and the clearfell step itself was not stationary, its early-era gain of +0.152 m falling to a late-era value indistinguishable from zero (Section 4.2.4) — so future BACI evaluations of topographic or canopy interventions should commit to multi-decadal baselines, lest a transient state observed in a short window be read as a permanent result. Second, because scraping is predicted to steepen the inland drainage gradient in the same way coastal erosion does, and the present network cannot resolve that drawdown, any future scraping should be paired with a dedicated transect of dipwells running uphill along the drainage gradient from the worked slack, so that the chase-inland acceleration can be measured directly and pre-empted rather than reconstructed after the fact.

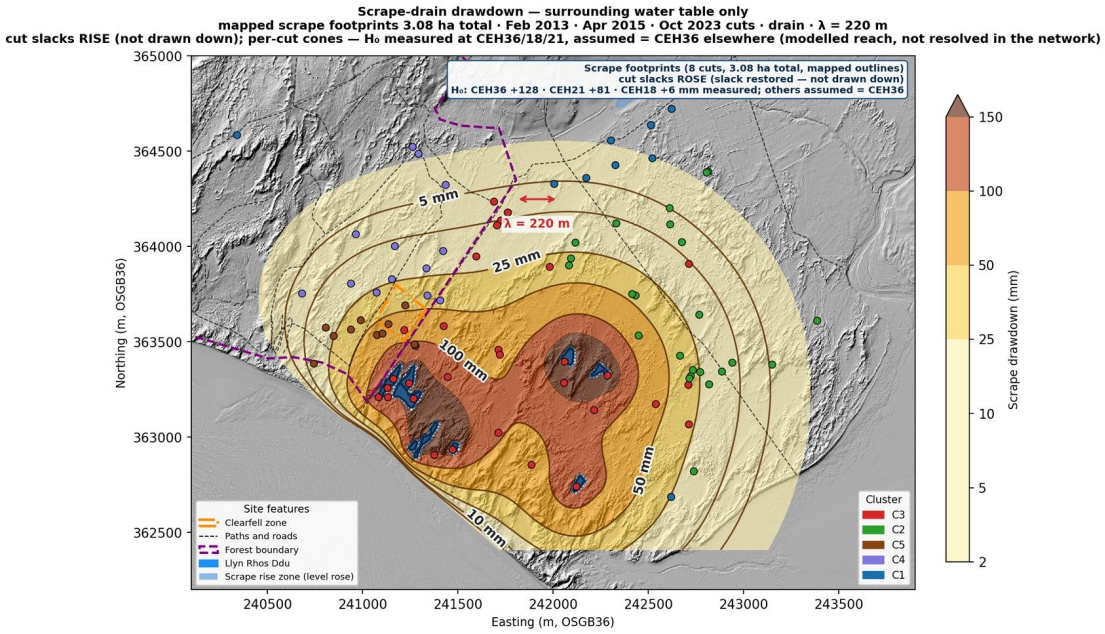


Figure 8. *Modelled steady-state drawdown field around the 2015 CEH36 scrape — edge magnitude empirically anchored, reach modelled. The edge drawdown is set to the measured CEH36 scraping response (≈0.13 m; Script 09a, Pure-Scraping era vs CEH4), not an assumed value; the surrounding field is a steady-state radial-drain decay along a flow-weighted cost-distance (leaky-aquifer length λ = √(K·b/(Sy·β₃)) ≈ 220 m), with the excavation interior masked because the slack itself rises. The excavation depth (≈0.43 m) is the inferred output, d_ex = H₀/Sy, not a survey input. In the model, the elevated displacement head at the scrape steepens drainage gradients and draws the surrounding water table down (not resolved in the record; Section 4.1.3) — the same mechanism as coastal erosion, underlying the chase-inland cascade. Source: 20_scrape_drawdown_nohead.*

Underlying all three is the binding constraint: the summer drought trajectory, under a climate redistributing rainfall toward wetter winters and drier summers, is approaching the SD15b and SD16 community-mean levels of Curreli et al. (2013) within the next one to two decades. This ordering is borne out when the interventions are placed on a common site-integrated footing with the site's background drivers (Hollingham, 2026d). Expressed as a local depth change, coastal-retreat drawdown is the largest spatially structured term, deepest at the dune toe and tapering inland over about 900 m. Integrated over the site the ordering inverts: an unexplained residual decline — the balanced observed decline less the modelled coastal gradient, averaged over the three open-dune clusters (−8.5 mm yr^{-1}) — is the largest single term because it is applied across the whole site, and coastal retreat is second. That term is a central estimate the record does not resolve, applied uniformly by assumption although the open-dune residuals range from about zero in the west (C3) to −14.6 mm yr^{-1} inland (C2), so the inversion is stated on that footing; the gradient fit's own constant is not used for it, being not separately identified from the cumulative-water-balance covariate. Set against the coastal term, the net scraping loss is about 25% and the clearfell gain about 7%. Climate is the leading candidate for the residual decline — the recharge-efficiency (β_1) record, the PET trend and the summer

temperature step point the same way — but the decomposition isolates what the coastal gradient leaves behind rather than a driver it has identified, and the β_1 decline across the BACI network is almost absent at the unfelled Climate-control wells, so its independence of management is not established (Hollingham, 2026d). The two management interventions are, on this measure, second-order against a background they cannot arrest. The management challenge at Newborough is therefore hydrological adaptation — maintaining flooding capacity in an increasingly seasonal climate — rather than restoration to a prior state. The prediction equations, threshold maps and scenario framework underpinning these conclusions are available as interactive tools (Hollingham, 2026e).

5.6 Transferability to other dune systems

Newborough is a particular site, but neither intervention tested here is particular to it. Removal of conifer plantation established on stabilized back dunes, and mechanical lowering of the ground surface, are the two commonest active interventions in dune-slack restoration across north-west Europe, and the results reported here bear on both wherever they are proposed. Three findings generalize; a fourth constrains how far this site's numbers should be carried.

The seasonal partition of the clearfell response is the first. That canopy removal was followed by a winter rise in the water table without a lift in the summer minimum is observed here at one felled well of a 4.2 ha compartment; it arose without the rise in recharge sensitivity or atmospheric draw that a sink-and-shield canopy would predict, and a felling wider than the drawdown reach could keep a recharge gain this one did not (Section 5.3). The pattern cannot be transferred without the site's own seasonal partitioning of the flooding-control mechanism, but the appraisal principle does transfer. Where the restoration objective is expressed as summer minimum depth — as it is in the eco-hydrological requirements of Curreli et al. (2013) and in most UK dune-slack designations — a clearfell should be appraised against a mean-level objective, quoting its winter gain (at the felled well, +0.128 m over October–March against −0.001 m over June–September; Hollingham, 2026d), and not credited in advance with a summer benefit. Ainsdale on the Sefton Coast, where tree cover and a drying climate act on slack water tables in the same combination, is the closest British parallel (Clarke and Sanitwong Na Ayutthaya, 2010).

The second is scraping as relief drainage. The mechanism set out in Section 5.1.2 — a scrape raises the displacement head at its own position and therefore drains its surroundings, in the same manner as an eroding shoreline — is a property of the geometry, not of this site. It offers a candidate explanation for a risk that management guidance already recognizes without explaining: that creating scrapes or pools can dry adjacent slacks (Buglife, 2019), a caution issued as a siting rule while scraping is otherwise treated as cut-to-the-water-table habitat creation (Grootjans et al., 2002; Denning et al., 2024). We are not aware of guidance that relates a scrape's geometry — its length, its orientation to the hydraulic gradient, and the base level to which it discharges — to its severity as a relief drain on the dune around it. Where scrapes are elongated and aligned with a coast-to-inland gradient, as they are here, that configuration is an efficient relief drain whatever its habitat intent, and the chase-inland cascade it can initiate is a design question anywhere the same geometry is used.

The third is the erosion analogue, which does not rest on this record alone. At Whiteford Burrows on the Gower coast of South Wales, the loss of some 4% of the spit width lowered the dune water table by approximately one metre (Robins et al., 2013). Direction and mechanism are what carry across, not rate: any dune system with an actively retreating margin should expect its interior water table to be falling for reasons management cannot influence, and should appraise its interventions on a footing that admits this.

What is most portable, finally, is the appraisal rather than any of the numbers. Reducing interventions and background drivers to a common site-integrated footing states a local gain against the site-wide cost that buys it (Hollingham, 2026d). On that footing both interventions are second-order at Newborough, against coastal-retreat drawdown, the largest local term, and an unexplained site-wide decline, the largest term integrated over the site; a sheltered or accreting system would rank differently, and it is the ranking exercise that transfers, not the ordering it produced here. The magnitudes do not transfer at all. The +0.108 m clearfell step is specific to one compartment, one control tier and one aquifer geometry, and is in any case measured net of the coastal confound rather than bounded by it (Section 5.2.1); the +0.195 m scraping benefit is a topographic step whose size is set by the excavation depth and the local drainage geometry. What another site can carry across is the appraisal framework and the drainage mechanism, with the seasonal partition as an observation to test locally — not the coefficients.

6. Limitations

Neither intervention was randomly assigned, raising the usual possibility that pre-existing site characteristics confound the signal; the monitoring network was not designed as a purpose-built propagation transect and cannot resolve a residual felling signal within 100 m of the clearfell edge. Both intervention analyses treat the coastal-erosion confound through covariates and control selection rather than through explicit time-lagged modelling: the coastal-drift term captures the average spatial gradient but not the multi-year propagation of erosion signals from earlier retreat, so the contemporaneous management signal cannot be fully separated from the time-integrated coastal signal. The coastal-drift covariate is a drift control, and the comparison against the physics-based gradient model is an internal consistency check rather than independent corroboration, the covariate being that model's field (Hollingham, 2026d). What survives as independent support for the coastal driver is the model-free coast-to-inland transect. The coastal driver is thus established in direction and corroborated in rate, but its precise amplitude cannot be independently recovered from the monitoring network alone (Hollingham, 2026d), which bounds how sharply the contemporaneous management and time-integrated coastal signals can be separated. The Thornthwaite PET method overestimates evaporative demand in humid maritime climates, so the absolute scenario magnitudes are conservative upper bounds; the relative ranking of scenarios and the finding of structural isolation are independent of the PET method because they depend on coefficient ratios and spatial structure. The NW10 broadleaf comparison rests on a single, positionally confounded well with fewer than four years of post-canopy-closure record and should be read as indicative only. The spatial framework is a per-well scenario aggregator interpolated by inverse-distance

weighting, not a calibrated continuous-flow model; resolving cluster-to-cluster coupling would require slug tests and a ground-penetrating-radar survey of aquifer thickness, identified as priority fieldwork. An empirical MSL5 change comparison (window end 2017 vs window end 2023) resolves no clearfell signal in five-year spring means; at the scraped well the benefit is absorbed by window overlap, the 2013–2017 baseline already containing post-scraping springs, although a paired comparison against CEH4 on the same metric retains it. Two-window differences depend on the windows chosen, because the amplitude of interannual spring variation differs widely across the network; that amplitude tracks β_2 rather than the recession time, so wells should be triaged on the interannual standard deviation of spring level rather than on $1/\beta_3$ (Hollingham, 2026d).

7. Conclusions

A 21-year manual record and a five-tier BACI design allow the hydrological effects of two dune-restoration interventions to be separated from a deteriorating climatic and coastal background. Against an in-situ forest control, the felled well rose +0.128 m over October–March after the December 2017 clearfell (95% CI +0.001 to +0.254 m, p = 0.048) and not at all over June–September (−0.001 m); the annual-mean step, +0.108 m, the record mean of a front-loaded recovery that was not sustained after 2020, is not significant on its own once autocorrelation is allowed for (p = 0.10), and summer minimum depth — the metric governing slack viability — was unchanged at every tier; the felled well shows no rise in recharge sensitivity and a fall in atmospheric draw, and whether felling acted through that reduced draw or through a recharge gain that a 4.2 ha compartment cannot keep, the record does not decide. The 2015 dune scraping delivered a larger and directly ecological benefit, improving the summer minimum at the scraped slack by +0.195 m, but did so by permanently lowering the ground surface — mechanically identical to coastal erosion, and carrying a chase-inland cascade risk that is modelled, integrated across the site, as a small net water-table loss; no off-site drawdown is resolved in the record. No felling response was detected outside the forest zone, and the scenarios, carrying no flow between wells, cannot show forest management reaching the eastern slacks where the ecological need is greatest. Summer minimum depth is the binding ecological constraint, best addressed by carefully sited scraping in the eastern transitional zone, while the binding constraint remains the climate-driven summer drought trajectory, to which a decline in recharge efficiency (β_1) across the BACI network is a candidate contributor, although it is almost absent at the unfelled Climate-control wells and its independence of management is not established. On a common site-integrated footing both management interventions are second-order: coastal-retreat drawdown is the largest local term, and an unexplained site-wide decline, carried as a central estimate the record does not resolve, is the largest term integrated over the site. The companion paper (Hollingham, 2026a) sets out the aquifer-characterization framework on which these conclusions rest.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Anthropic's Claude (successive Claude Opus and Claude Fable models, accessed through the Claude desktop application) for substantial drafting and editing of the manuscript text: composing prose from the author's analytical outputs and decisions, structuring and restructuring sections, and correcting the text against the study's committed analytical outputs. The conception and design of the study, the twenty-one-year manual dipwell record on which it rests, the analytical decisions, and the interpretation of the results are the author's own. Numerical values in the manuscript were verified against the committed outputs of the analysis pipeline, and a decision log records the methodological and editorial calls made by the author during the work. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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